

EXAMPLE USE CASES, OPERATIONAL SCENARIOS, AND NON-LIMITING APPLICATIONS

General Operational Scenarios

The longitudinal penile physiology monitoring device may be used in a plurality of operational scenarios including continuous day wear, sleep monitoring, episodic physiologic activation tracking, activity-associated monitoring, recovery trend observation, and longitudinal baseline establishment. Monitoring may occur in a continuous manner or through periodic sampling depending on selected mode and user configuration.

Sleep and Nocturnal Monitoring Scenario

In certain embodiments, the device may be worn during sleep to identify nocturnal erection events and characterize frequency, timing distribution, duration, rigidity level, and stability of sleep-associated physiologic episodes. Inertial sensing may identify low-motion windows to improve confidence of nocturnal event detection and support contextual interpretation of physiologic transitions during rest cycles.

Performance and Episodic Monitoring Scenario

In certain embodiments, the device may operate in high-fidelity or performance mode during selected periods to capture detailed physiologic dynamics including erection onset speed, rigidity rise slope, peak stability behavior, detumescence pattern characterization, and context-correlated perfusion or temperature responses. Performance monitoring may be user-initiated, scheduled, or automatically triggered upon detection of physiologic transitions.

Longitudinal Trends and Baseline Establishment

Over extended monitoring periods, the device may establish individualized baselines for rigidity behavior, episode frequency, stability metrics, and recovery patterns. Longitudinal trends may be used to detect changes over time, compare present behavior to prior intervals, and generate personalized physiologic profiles. Baseline models may adapt to user variability and support normalization across environmental conditions.

Integration and Cross-Node Applications

In multi-node configurations, the device may operate in conjunction with additional sensing nodes to enable cross-device correlation, comparative physiologic evaluation, and composite insight generation. Such applications may include synchronization with cardiovascular monitoring nodes, internal physiologic monitoring nodes, structural mapping nodes, or partner-associated nodes when permitted by configuration and privacy controls.

Non-Limiting Application Statement

The scenarios described herein are non-limiting examples. The disclosed device may be applied to any use case in which longitudinal characterization of penile physiology, rigidity dynamics, contextual physiologic behavior, or multi-modal wearable sensing provides value. Variations in mode selection, duration of wear, sensing configuration, and computational interpretation are encompassed within the scope of the invention.